Round 16 | Quartic memory

One carried state, tested before compression
Astra research note | 8 October 2026

No qualifying state established. I tested a dense pair-innovation core born at every ReLU, with exact transport over a six-age window. Its decisive-neuron pattern correlations at L12-16 are **0.369-0.435 [M]**, below **0.7 [Q]**. Keeping every birth gives **0.362-0.433 [M]**. Reject this mechanism; extending its horizon does not fix the pattern.

Labels: [M] measured in the attached scripts; [D] derived, including totals extrapolated from measured kernels; [Q] quoted from your prompt or project; [E] estimated. All equation constants, dimensions and index ranges are [D] unless marked. No accuracy estimate is substituted for a measurement.

State and birth rule

Use the row convention $z_{l+1} = h_l W_l$. At each birth b keep a dense n by n core G_b and reference the chain's $P_{b,l}$ propagator. Let $d_b = \mathrm{Phi}(mu_b/sigma_b)$. Superscripts + and - below mean the baseline chain's post-ReLU slices and regenerated pre-ReLU slices, respectively.

```
Delta4_a = K4+_a - d_a^4 K4-_a
Delta22_ab = K22+_ab - d_a^2 d_b^2 K22-_ab
(G_b)_aa = Delta4_a
(G_b)_ab = 3 Delta22_ab,  a != b
```

Use the symmetric part of G if stored roundoff breaks symmetry. At the first birth the inherited cumulants are zero. These are **baseline-defined innovations**, not the true nonlinear residual of a full fourth-order tensor. The birth approximation retains the (4) and (2,2) sectors; it omits newborn (3,1), (2,1,1) and (1,1,1,1) content.

Transport and diagonal readout

```
P_(b,b+1) = W_b.T
P_(b,l+1) = W_l.T diag(d_l) P_(b,l)
I_l = {b: max(1,l-6) <= b <= l-1}
S_b = P_(b,l) * P_(b,l)
q_i = sum_(b in I_l) sum_a [(S_b G_b)_ia (S_b)_ia]
k4hat = q * mean(k4chain) / mean(q)
```

The asterisk is elementwise multiplication. Means in the last line run over all neurons, preserving the raw per-layer diagonal level. Apply the correction only at L8-16 [D, chosen schedule]; earlier mean closures stay at their baseline. A source can cross up to six linear maps, including its birth-to-next-layer map [D].

Size and integration

At n = 1024 [Q], six active cores need **24 MiB [D]** in float32. For simple chronological slices, the costed implementation allocates all fifteen birth slots: **60 MiB [D]**, plus **48 MiB [D]** for S and SG. Existing dense P views are reused. The ceiling test instead reconstructs uncompressed P in float64; the current graded chain's shared-basis approximation is not part of that test.

The falsifier was run

Public MLP #0 | exact contraction of the chosen state | no teacher feedback

Using the stored chain slopes and birth slices, I computed the complete dense G and P matrices, with no rank truncation, basis restriction, source screening or learned coefficients. Teacher arrays enter only the score. The second run removes the age cutoff while keeping the same birth rule and readout; it is a control of the same mechanism.

```
mask_i = |mu_true_i / sqrt(v_true_i)| < 1.5
rho = corr(k4hat / v_true^2, K4_true / v_true^2) on mask
```

The cutoff 1.5 and required correlation 0.7 are [Q]. This standardized-kurtosis convention reproduces the project's low baseline correlations. Raw-diagonal correlations are also shown to expose the distinction. Positive mean-level rescaling does not change either correlation [D].

Layer	Decisive neurons	Baseline pattern	Six-age pattern	All births pattern	Raw corr. six / all
12	442	0.339	0.435	0.433	0.694 / 0.743
13	406	0.281	0.391	0.389	0.649 / 0.716
14	357	0.298	0.401	0.390	0.664 / 0.727
15	375	0.264	0.369	0.362	0.633 / 0.728
16	323	0.324	0.426	0.411	0.615 / 0.708

All table values are [M]; layer numbers are indices. Neither schedule reaches the gate at any tested deep layer. The baseline is the older dense capture chain, not a fresh run of graded V55. No compressed rollout, synthetic-network replication or final-MSE claim is warranted by this failed ceiling.

Checks on orientation, normalization and contraction

P reconstruction errors against captured legs are at most **6.59×10^{-7} for the six-age run and 1.05×10^{-6} for all births [M]**. Stored Wick slopes match Phi(mu/sigma) within **4.78×10^{-8} absolute error [M]**. On an explicitly materialized symmetric tensor of width **5 [D, check size]**, the readout agrees to **1.52×10^{-16} relative error [M]**.

Thus the large-width contraction is algebraically exact for this chosen quartic state. Materializing its n by n by n by n array would give the same result; it would add storage, not information. This is an uncompressed-state ceiling test, **not an oracle over all possible birth cores**. The source captures themselves contain some float32 inputs, which were promoted to float64.

Does it help the mean-closure channel?

```
g_i = phi(alpha_i)(alpha_i^2 - 1)/(24 sigma_i^3)
R = ||g * (k4hat - K4_true)|| / ||g * (k4chain - K4_true)||
```

Across L12-16, R is **1.119-1.427 [M]** for the six-age state and **1.012-1.151 [M]** with all births. This frozen-state, teacher-lever diagnostic is an error-norm ratio, not final-layer MSE. Its failure reinforces the correlation verdict. The two full-width ceiling runs took **23.6 s and 30.1 s [M]**, respectively, with single-threaded BLAS on this workstation.

Actual cost at the real shapes

Meter: flopscope 0.12.1+np2.2.6 [M] | current V55 _Strassen class | float32

Assume the chain supplies P and the pre/post birth slices. Each corrected layer uses **six (1024 by 1024) products [D]**, batched as (6, 1, 1024, 1024), for $S_b G_b$. There are **nine readouts and fifty-four square products [D]** over L8-16. This is six n-cubed products per readout, not an n-squared contraction.

Kernel	Exact FLOPs [M]	Ops [M]
First Gaussian-input birth	1,049,600	2
Each later birth	4,199,424	9
Six-source readout, level 0, warm	12,897,486,849	7
Six-source readout, level 3, warm	8,818,165,761	73
Six-source readout, level 4, warm	7,865,624,577	97
Six-source readout, level 5, warm	7,176,507,393	119

Readout counts include squaring P, the matrix products, contraction over source and hub indices, and mean-level rescaling. Birth counts include the diagonal overwrite. Dense products use the meter's multiply-add convention; allocation/page-touch charges are treated separately below. Random arrays were used to meter shapes; this was not a cheap-version accuracy test.

Multiply level	Whole network: warm pool u / ops [D]	Whole network: dedicated pool u / ops [D]
0	54.080609 / 194	54.080609 / 194
3	36.984372 / 788	37.321469 / 820
4	32.992318 / 1,004	33.328854 / 1,041
5	30.104261 / 1,202	30.734215 / 1,247

```
F_warm = F_birth1 + 14 F_birth + 9 F_query
O_warm = 3 + O_birth1 + 14 O_birth + 9 O_query
```

The extra three operations allocate G, S and SG [M]. The dedicated-pool column replaces the first warm readout by a cold one. At level 5 the cold readout is **8,529,324,033 FLOPs / 164 ops [M]**; the resulting whole-network cost is **66,001,224,713 FLOPs [D]**. With an already compatible pool it is **64,648,408,073 FLOPs [D]**. One u = 2^{31} FLOPs [Q].

What this ledger does and does not buy

The ordinary separate readout schedule is near the **30 u [Q]** allowance but exceeds the **1,000-op [Q]** target. Kernel fusion might reduce operations; these counts are not a lower bound. The level-5 pool occupies **1,397,071,872 bytes [M]**, excluding the existing chain. Combined peak memory was not benchmarked.

An independent six-age exact P carrier adds **60 square products plus 60n^2 slope multiplies [D]** to this schedule. Fully classical standalone arithmetic is **114.080609 u [D]**. Retaining every birth uses **99 readout products [D]**; with reused P and warm level-5 buffers its readout-and-birth subtotal is **55.167916 u [D]**. Both versions failed accuracy, so neither is a passing fallback.

Why this is a useful rejection

The missing ingredient is not established by long memory alone

The quartic state really has all-distinct entries

```
F_(b,l)(v) = [(P_(b,l).T v)^2].T G_b [(P_(b,l).T v)^2]
K_(b,l)[v,v,v,v] = F_(b,l)(v)
```

Squares here are elementwise. At birth, the coefficient of $v_a^2 v_b^2$ is $2G_{ab}$ = 6 $Delta22_{ab}$ [D], fixing the factor of three. After dense transport, expansion generates both (2,1,1) and (1,1,1,1) entries. Their coefficients depend on signed mixed paths in P. Squaring individual weights before transport would lose those terms.

However, generating nonzero all-distinct entries is not equivalent to generating the correct ones. The birth restriction and slope-only nonlinear evolution remain approximations. The full-history failure shows that simply retaining this core longer does not repair them; the experiment does not isolate which omitted birth sector is responsible.

Why this was not already killed by the listed tests

Round 13's pair-core experiment set later births to zero [Q, prior report]. This test adds a dense innovation at every ReLU. The previous young-window result used a shorter horizon [Q]; here the six-age run is accompanied by the complete source history. A carried pair-response mode keeps a small family of propagated matrices [Q, project probe]; this state keeps a full hub-pair coefficient matrix per birth. Reusing P does not make G a cubic readout: its coefficients come from fourth-cumulant slices, not source-resolved skew energies.

Two limits on the motivating oracle arguments

Parity [D]. For a fixed upstream tensor, a fourth-order contraction is unchanged by a global W to -W sign flip: it has four weight factors. The cubic odd-parity theorem therefore does not transfer to this object. Individual-coordinate sign flips can still change mixed terms. The measured failures of cheap even-statistic predictors [Q] remain useful evidence, but are not a global odd-parity theorem for this quartic carry.

The history oracle [D/Q]. A Gaussian restart followed by actual ReLUs regenerates and mixes cumulants of several orders. Its quoted six-layer correlation of 0.77 [Q] is not a ceiling guarantee for linear transport of a restricted quartic birth. Similarly, the true one-step full-tensor oracle continually starts from information this carrier has discarded.

Reproduce and decision

The PDF embeds **ceiling_probe.py**, **ceiling_result.json**, **full_history_result.json**, **meter_probe.py** and **meter_result.json**. Extract them into a writable folder. Run the commands below; the meter script needs the project virtual environment. The scripts read the project and write diagnostic outputs beside themselves. No project source was edited and no commit was made.

```
python ceiling_probe.py
python ceiling_probe.py --full
python meter_probe.py
```

Inputs: `data/dumps/r7_cap_L{l}.npz`, `errbudget_v29plain.npz`, and public pre-teachers/weights under `data/kaggle_out/kaggle_out_k4pub/k4teach`. All required birth fields were available; full P_st captures were found only at L3/7/11/14/15 [M] and used as cross-checks. Meter implementation: `submissions/v55k4c/estimator.py`. Background reference: ARC's public cumulant-propagation code; it supplies no accuracy claim for this proposed state.

Decision: stop this pair-innovation carrier before compression or deployment. I have not found a state meeting your accuracy gate, and I cannot honestly name a cheapest passing fallback from these measurements. This is a rejection of the specified mechanism, not an impossibility proof for cheap quartic states.